Pricing

MGT 100 Week 6

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FTC Surveillance Pricing Study Indicates Wide Range of Personal Data Used to Set Individualized Consumer Prices

The agency details interim insights from staff perspective examining how companies track consumer behaviors to inform surveillance pricing

January 17, 2025

The staff perspective is based on an examination of documents obtained by FTC staff's 6(b) orders sent to several companies in July aiming to better understand the shadowy market that third-party intermediaries use to set individualized prices for products and services based on consumers' characteristics and behaviors, like location, demographics, browsing patterns and shopping history.

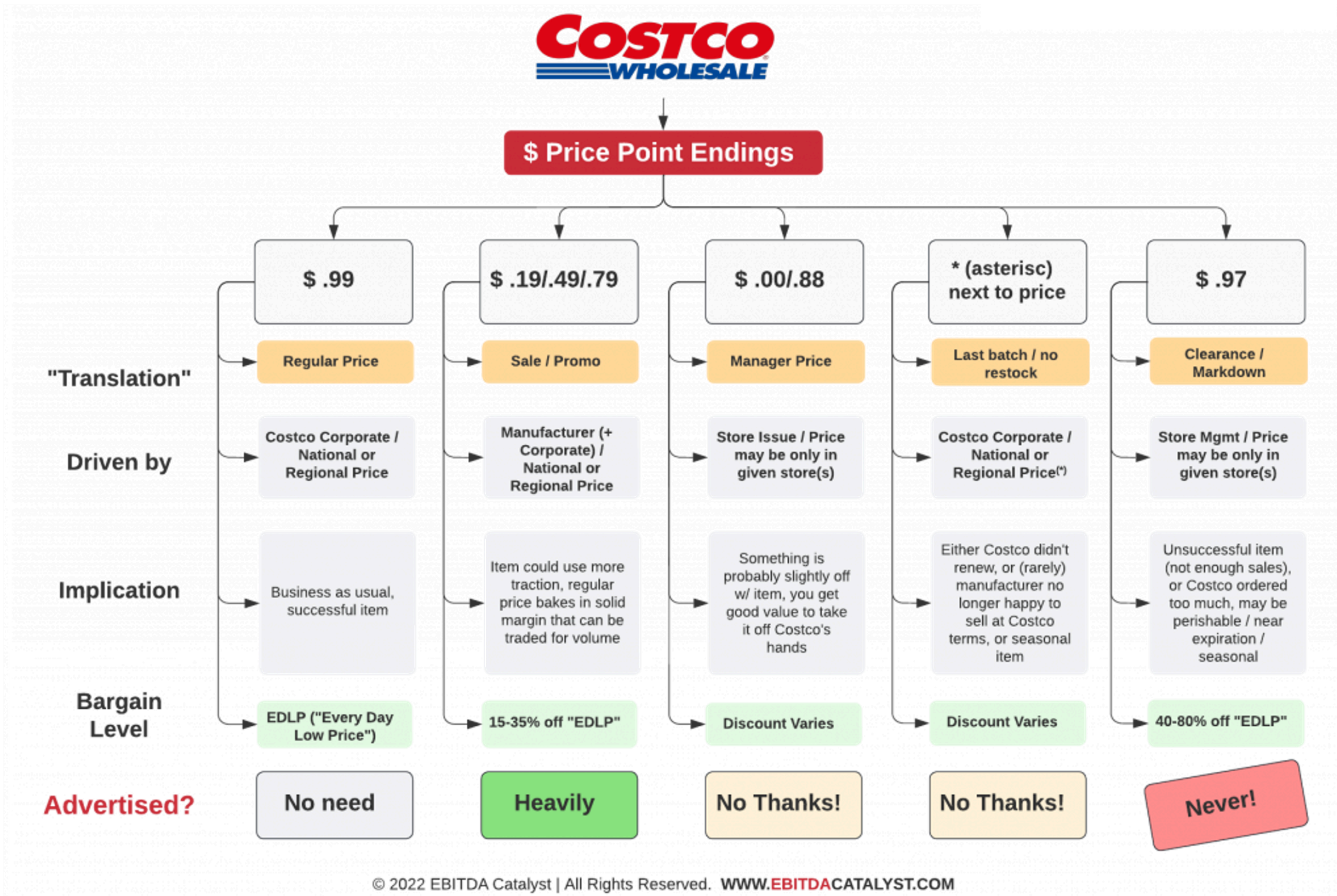
Staff found that consumer behaviors ranging from mouse movements on a webpage to the type of products that consumers leave unpurchased in an online shopping cart can be tracked and used by retailers to tailor consumer pricing.

"Initial staff findings show that retailers frequently use people's personal information to set targeted, tailored prices for goods and services—from a person's location and demographics, down to their mouse movements on a webpage," said FTC Chair Lina M. Khan. "The FTC should continue to investigate surveillance pricing practices because Americans deserve to know how their private data is being used to set the prices they pay and whether firms are charging different people different prices for the same good or service."

The FTC's study of the 6(b) documents is still ongoing. The staff perspective is based on an initial analysis of documents provided by Mastercard, Accenture, PROS, Bloomreach, Revionics and McKinsey & Co.

> As far as I know, most companies are not using sophisticated personalized pricing algorithms. It would be possible for them to do so. To me, this speaks to a widespread concern about the nontransparency of personalized pricing practices, and the possible use of personal data to work against customer interests.

FTC (2025)

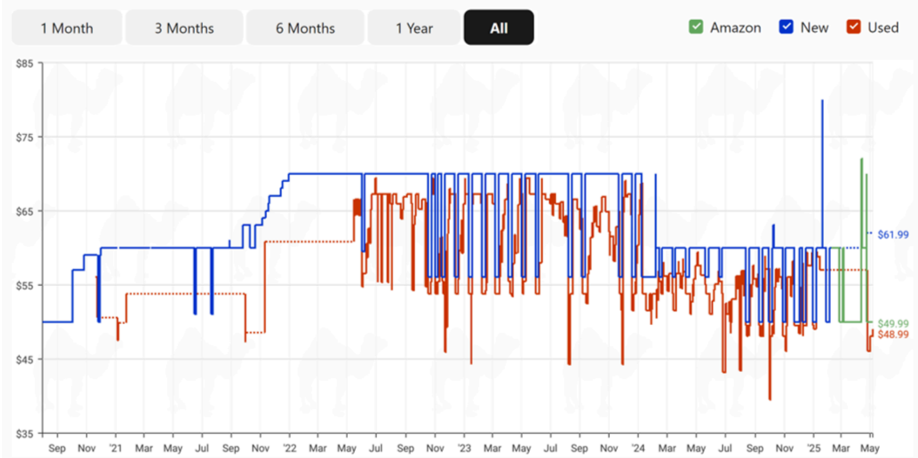


> Costco practices indicate the value and versatility of using price as a signaling tool. But why do you think they use price, rather than just putting signs on the shelf?

EBITDA Catalyst

Amazon Price History

This is our record of Amazon price changes for **HBTower Step Ladder, Folding Step Stool with Wide Anti-Slip Pedal, Sturdy Steel Ladder, Convenient Handrail, Lightweight, Portable Steel Step Stool (B087279HM1)** since we began monitoring it on on Aug 01, 2020.



Price Type	Lowest Ever	Highest Ever	Current +	Average
Amazon	$49.99 (Feb 24, 2025)	$71.96 (Apr 12, 2025)	$49.99 (Apr 22, 2025)	$59.09
3rd Party New	$49.99 (Nov 24, 2020)	$79.99 (Jan 20, 2025)	-	$59.21
3rd Party Used	$39.49 (Oct 01, 2024)	$69.29 (Oct 28, 2022)	$48.99 (May 05, 2025)	$52.48

Camelcamelcamel and other price trackers can help you decide whether to buy now or wait. How many shoppers use them?

camelcamelcamel | See also PriceSpy, SmartScout; but not Honey or Keepa

How firms set prices

- Importance and challenge
- Common approaches
- Economic Value to the Customer
- Van Westendorp

Pricing importance

- #2 topic after two-sided value creation
- Widely cited research by McKinsey
    - → 1% price increase can lead to 11% profit increase
    - → 1% price decrease can lead to 8% profit decrease
    - → Logic assumes no decrease in quantity
    - → Correlations, but widely misinterpreted as causal
- Consultants say most companies price too low; price is low-hanging fruit
- Counterpoint: “Your margin is my opportunity”

> Competition usually leads to thin margins. 5-15% is common. Therefore, minor pricing improvements can lead to substantial profit improvements.

Price changes are risky & scary

Harvard Business School

Note on Behavioral Pricing

Unfortunately, it appears that many firms lack the necessary understanding of a consumer's "willingness to pay" to optimally set product prices. When asked whether they were "well-informed" on six of the potential inputs to the product pricing decision, managers at one well-respected U.S.-based multinational responded as follows:[2]

- 84% were well-informed on the variable cost of providing their product.
- 81% were well-informed on the fixed cost of providing their product.
- 75% were well-informed on the price of competitors' products.
- 61% were well-informed on the value of their product to the customer.
- 34% were well-informed on how consumers would respond to price changes.
- 21% were well-informed on consumers' willingness to pay at various price levels.

These managers were well-informed on the costs of providing its products and on the price of competitor's products. They were also well-informed on the value its products delivered to consumers. However, when it came to a consumer's willingness to pay or to a consumer's response to potential price changes, these managers were lacking the insight needed to optimally set prices. Experience suggests that this company is not unique in this regard.

> Internal costs are easiest to discover. Competitor actions are the second-easiest to track and know. Customer factors are usually the biggest gap. Why do you think that is?

How firms usually set prices

- Most common: Limited-data analyses
  - → EVC / Value pricing, Competitor price benchmarking, Cost-based pricing
- Stated-Preference Data
  - → Open-ended: How much are you wtp? $___
  - → Prompted: Would you buy (product) at ($price)?
  - → Interviews, Focus Groups, Van Westendorp surveys
  - → Conjoint Analysis: Designs can be incentivized or not
- Revealed-Preference Data
  - → Simulated purchase environments, Test markets
  - → Algorithms (bandits, rev mgmt), Experiments (Amazon pricing labs)
  - → Demand estimation
    - Requires data, exogenous price variation, human attention/expertise
- Customer co-determination
  - → Monopsony, auctions, negotiation, pay-what-you-want
- None: Seller takes market price

Pricing strategies are secret

- Why not announce your pricing strategy?
- We can survey pricing managers anonymously, but (i) nonrandom selection, (ii) survey design inconsistencies, and (iii) self-reporting biases
- Salt taken, let's look at pricing manager surveys

> What are the costs and benefits of price strategy transparency?

Table 3: How is the Price of Your Main Product Set

Firm Size and Sector

Firm size	Micro 5-9	Small 10-49	Medium, 50-249	Large 250+
No autonomous price setting	8.6	15.9	14.1	25.9
Price set by customer(s)	6.0	4.2	4.3	9.4
Price set following main competitors	34.0	32.3	30.7	30.9
Price based on costs and self-determined profit margin	45.6	41.3	42.6	30.1
Other	5.8	6.3	8.3	3.7
Sector	**Manufacturing**	**Construction**	**Trade & Distribution**	**Other services**
No autonomous price setting	9.7	4.5	18.3	7.6
Price set by customer(s)	9.8	7.3	3.9	5.1
Price set following main competitors	22.5	27.9	32.7	37.6
Price based on costs and self-determined profit margin	57.6	56.2	39.9	41.6
Other	0.4	4.1	5.2	8.1

Cost-driven pricing and Competitor price-tracking were the two most common approaches, by far. Why?

Keeney, Lawless & Murphy (2010) | 1,000 Irish businesses surveyed — how often is each approach used?

> Digital-native businesses are more savvy, on average. Why do you think Value-based pricing correlates so strongly with annual revenue?

OpenView (2023) | Survey of SaaS pricing managers, by ann. rev. | Top 3 strategies so common that others are unreported

Table A1: Pricing practices by firms with and without pricing algorithms (Survey)

	Pricing Algorithms used	
How often does your firm change prices?	No	Yes
1= continuous	0.037	0.130
2 = hourly		
3 = daily	0.074	0.148
4 = weekly	0.074	0.074
5 = monthly	0.000	0.148
6 = quarterly	0.296	0.222
7 = yearly	0.333	0.167
8 = less than yearly	0.111	0.074
Customize		
Individualized to each customer	0.316	0.243
Customized geographically and each consumer segment	0.368	0.432
Customized for different segments	0.105	0.054
Customized for different regions	0.211	0.162
Uniform across segments and geographic regions	0.000	0.108
varies prices across customers and geographies		
Individual to each consumer;	0.259	0.167
Geographically and each consumer segment	0.333	0.352
Customized for different segments	0.111	0.130
Customized for different regions	0.111	0.167
Uniform across segments and geographic regions	0.000	0.111
Different parts of the company use different price variations	0.185	0.074

Table A2: Data and type of method used for pricing algorithm (Survey)

Data used for Pricing Algorithm (most familiar product)	**Proportion**
Your firm's costs	0.757
Your firm's past revenue or profit data	0.730
Competing firm's prices	0.568
Past consumer behavior	0.541
Demographics and Geographics	0.595
External info	0.405
Type of Pricing Algorithm (method or rules used)	**Proportion**
"Win-Continue Lose-Reverse" rule	0.297
Q-Learning	0.135
Artificial neural networks	0.054
Deep learning	0.135
Adaptive machine learning	0.270
Unsupervised or reinforcement learning	0.108

> This survey sample came from a platform promoting pricing algorithms; the bias is clear in the results. But even still, many firms using algorithms do not update their prices very often. Why?

Spann et al. (2024) survey of pricing managers on EPP, a pricing/customer-growth nfp promoting algorithms

Value Pricing: Price in (cost, wtp)

- But… how do you learn wtp? Esp. if you have not sold before?
  → For large time/budget: Conjoint, simulated purchase environments, test markets, …
  → For small time/budget: Economic Value to the Customer (EVC)
- EVC: estimates customer benefit from a product, relative to the next best alternative
- EVC & VP are often used by new firms, highly differentiated products, firms lacking credible market research and related expertise
- Steps: 1. Calculate EVC, 2. Choose a price in (Cost, EVC)

> Some of you will start companies, and you will use EVC to identify value-improvement opportunities, investment potential, and price ceiling; and you will likely recalculate EVC many times as you refine your offering

How to calculate EVC(x|y)

1. Select the best available alternative y and find its price
    - Interview target customers to learn how they solve the core need (y)
    - If wrong y, EVC estimate will be too high
2. Determine non-price costs of using y and x
    - Include start-up costs and/or post-purchase costs
    - Make sure NonPriceCosts(x) exclude the price of x (why?)
3. Determine the incremental economic value of x over y
    - Usually, functional benefits or non-price cost savings
4. EVC(x|y) = Price(y) + (NonPriceCosts(y) - NonPriceCosts(x)) + IncrementalValue(x|y)
    - In practice, 99% of effort is getting the assumptions right

EVC tips

- y might not be a commercial product
- EVC and y often vary across customer segments
  - → Calculate heterogeneous EVC(x|y) for multiple y
- Qualitative factors influence price selection in (cost,EVC)
- If EVC(x|y)<0, reconsider target customer and/or value proposition

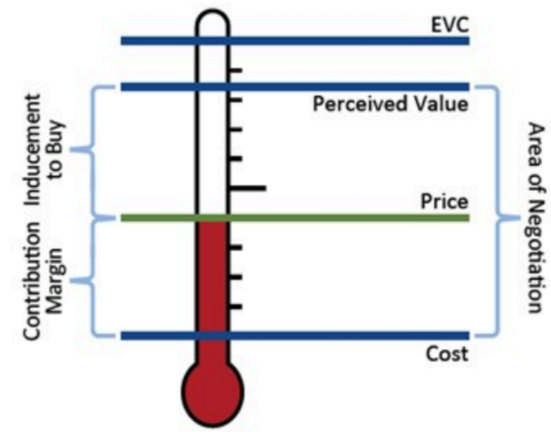
Example: What is EVC(Batteroo Boost|y)?

- The Batteroo Boost is a durable metal sleeve that increases disposable battery life by 800%. With a thickness of just 0.1 millimetres, the sleeve can be fitted over any size battery, in any size compartment
- Assume the typical battery costs $0.50

> Batteroo Boost raised almost $400k on Indiegogo

Choosing p in (cost, EVC)

- Pricing 'Thermometer:' How much inducement do you give your customer?
- Some advise: $Price = Cost + (EVC - Cost) \cdot z\%$
  - → I've heard z = 25%, 33%, 50%, and 70%
  - → Do you want profits or growth?
- Human factors to consider:
  - → Perceived benefit - actual benefit
  - → Perceived costs - actual costs
  - → Consumer price sensitivity, reference price
  - → Established pricing benchmarks
  - → Fairness, signaling
  - → Customer risk of adoption, skepticism; brand credibility



> Why do you think EVC exceeds Perceived Value?

Why not just ask customers for their WTP?

- Hypothetical bias: Stated answers ≠ real behavior
- They don't know: WTP is constructed in context, not stored in memory
- Strategic bias: Respondents may overstate or understate to influence firm's product launch decision or pricing decision
- Lack of competition: Purchases require tradeoffs vs. alternatives
- Social desirability/Identity: People may signal self-image ("I'm not cheap")

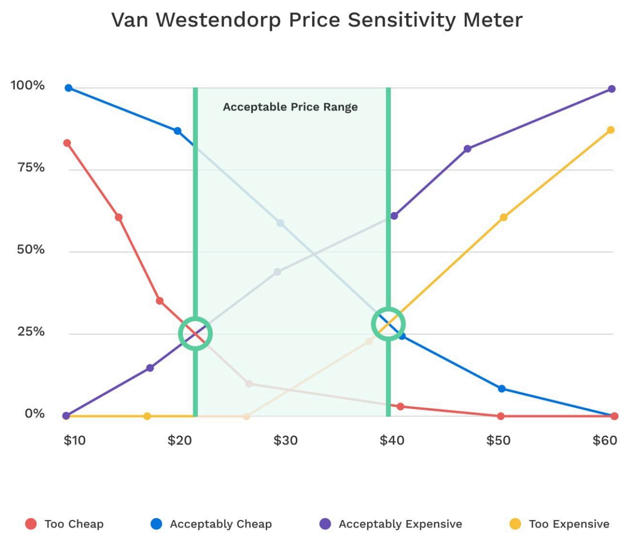
> Miller et al. (2011) find that incentivized methods recover true demand curve better than non-incentivized methods; non-incentivized methods underestimate price sensitivity

Miller et al. (2011)

Van Westendorp Pricing Model

- Goal: Elicit distributions of customer price perceptions
- Survey target customers: At price $X is (product)…
  - → Too Cheap? I.e., that you would question its quality
  - → Acceptably Cheap?
  - → Acceptably Expensive?
  - → Too Expensive? I.e., that you would not consider buying
- Ask for many values of $X, then plot 4 CDFs
  - → Too Cheap and Acceptably Cheap decrease with price
  - → Acceptably Expensive and Too Expensive increase with price
  - → Crossing points bound the “Acceptable Price Range”

R package pricesensitivitymeter



- “Too cheap” meets “Acceptably Expensive”: “Point of Marginal Cheapness”
  - → VW says: Price<PMC signals poor quality
- “Acceptably Cheap” meets “Too Expensive”: “Point of Marginal Expensiveness”
  - → VW says: Price>PME prices out most of your market
- “Too Cheap” meets “Too Expensive”: Possibly min. # of price-refusers
- “Acceptably Cheap” meets “Acceptably Expensive”: Possibly max. # of price-accepters
- These confusing ideas are largely hypothetical and not well validated

Van Westendorp (1976)

Pricing factors

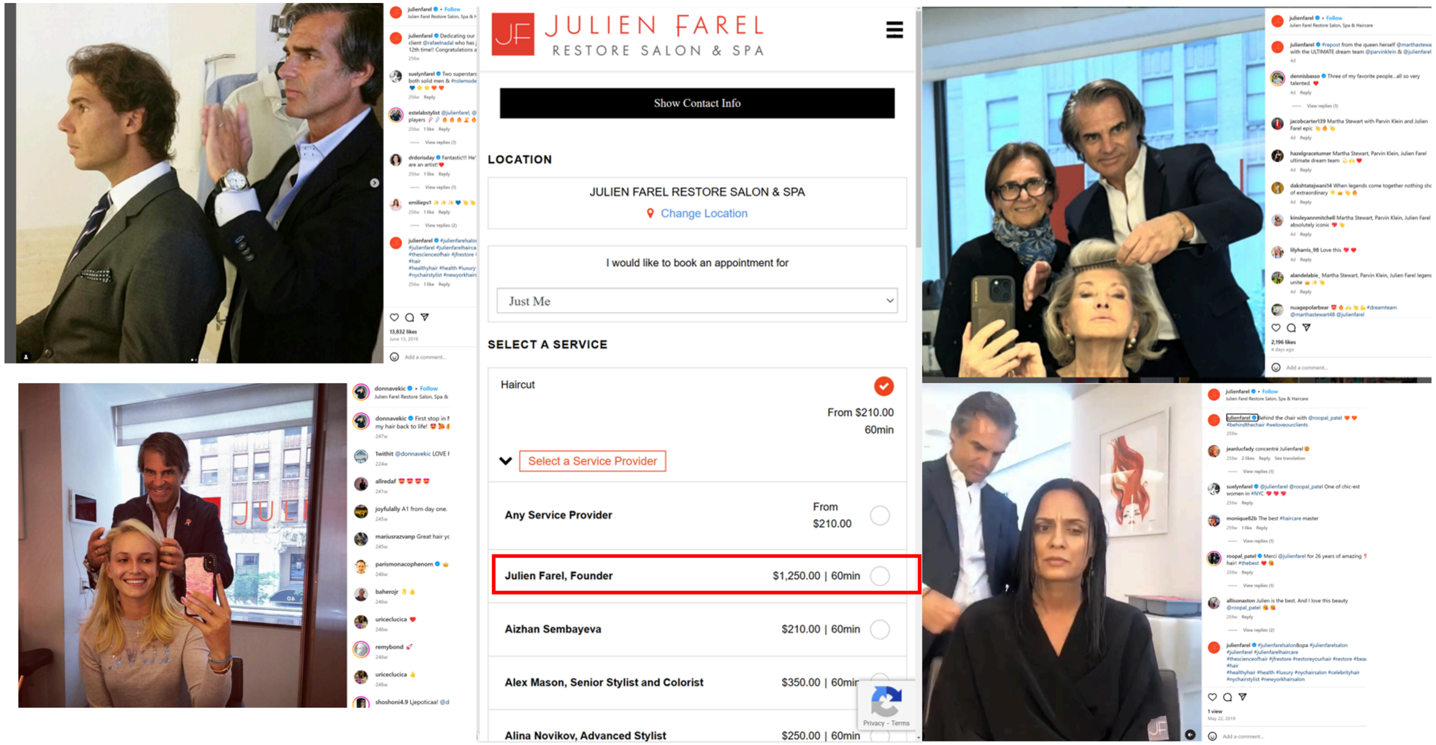
- Human Factors
- Economic Factors
- Price Elasticity of Demand

Signals and *Perceived Quality*

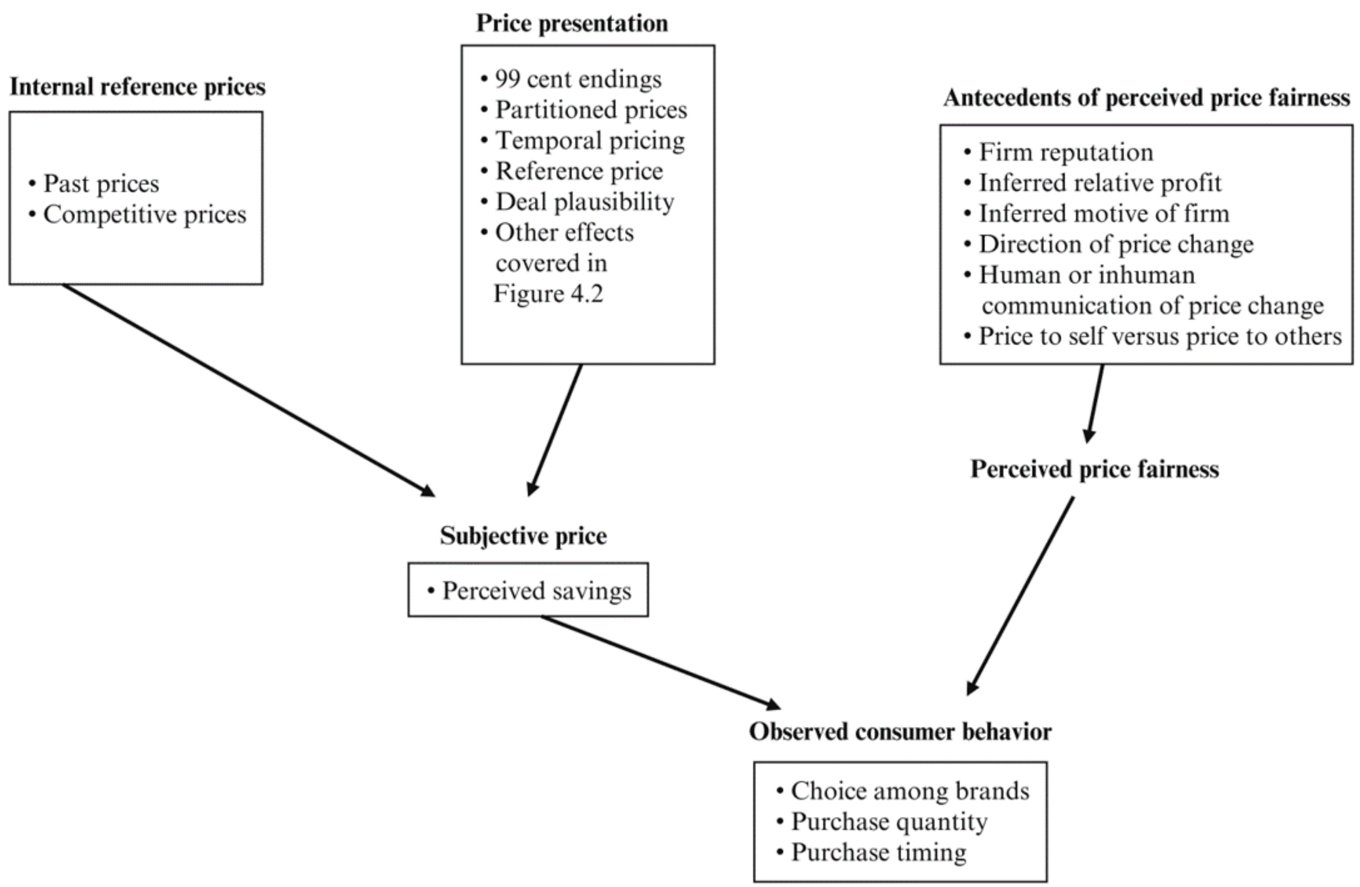
- Signals of high quality
  - → High prices, Brand names, Warranties, Return policies, Ad spending (e.g., Super Bowl ads)
  - → Costly signals when the firm doesn't deliver
  - → Brand reputation can convey credibility
- Signals of low quality
  - → Low prices, Price promotions, Price-matching guarantees
  - → Low-quality ads, or too many ads, "B-list" celebrities
  - → Signals that look too good to be true
  - → "If it's so good, why is it so cheap?"
- Prescription: Price consistent with your quality position in the market
  - → Otherwise, you undercut your own message and leave money on the table
  - → Findings replicate in numerous contexts

> Suppose you offer the best product in the market at the lowest price; will you be able to corner the market?

Human factors: Price as a signal



> Price can be a powerful signal of quality. Occasionally, we even see reverse-price-wars. How much do you think Nadal paid for that haircut?

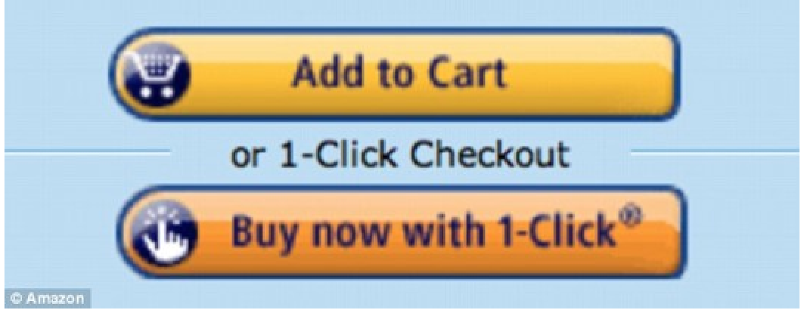


> Perceived price is affected by objective price, price presentation, and perceived price fairness, which in turn is driven by consumer factors like experiences, income, reference price, and comparison to others. MGT 104 & 107 go deeper

Krishna (2009)

Human factors: Cognitive Costs

- Total customer cost is
  = Cognitive cost to decide the purchase
  + Physical cost to acquire the product
  + Financial payment
- Simplicity can increase sales. Remove frictions



> Have you ever abandoned a purchase because you had to jump through too many hoops? 72% of e-commerce carts get abandoned without purchase. Counterpoint: A limited amount of friction can help sellers of complex products to screen customers, engage customers, and communicate with customers

Bertini et al. (2023)

Human factors: Perceived prices

- Which is the better bargain?
  - → Regular price \$0.89, sale price \$0.75
  - → Regular price \$0.93, sale price \$0.79

> How do you evaluate these two pairs of price discounts? How do most consumers evaluate them?

Left-digit bias: Demand Effects

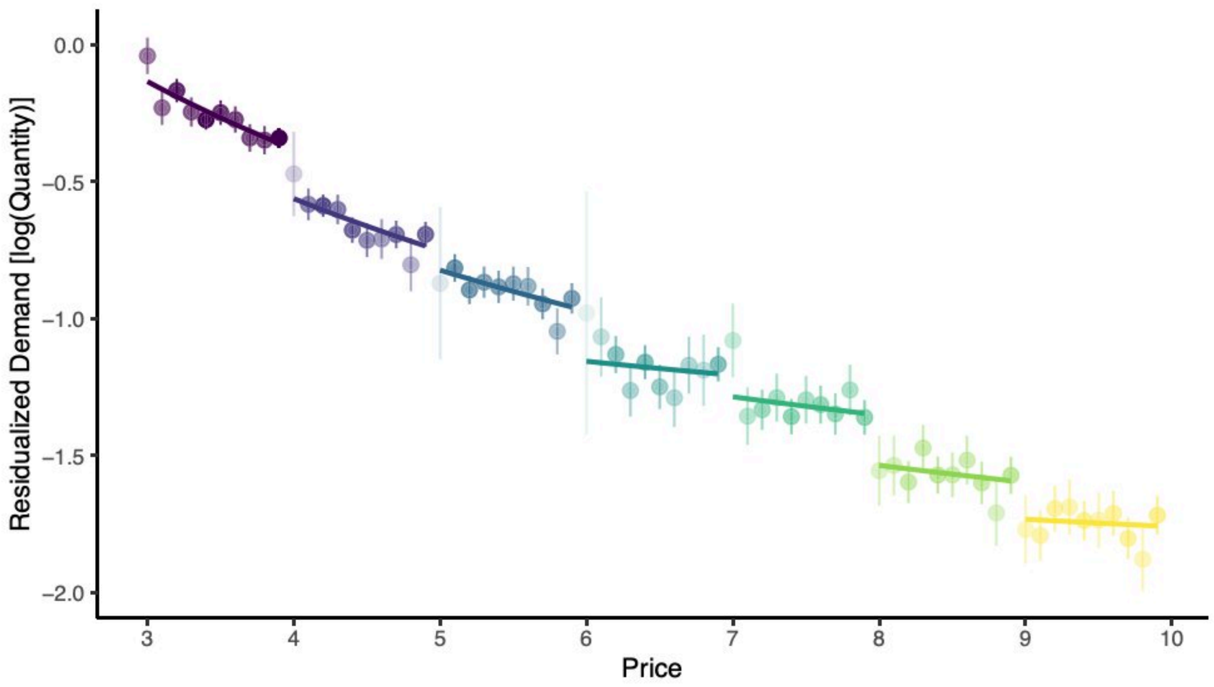
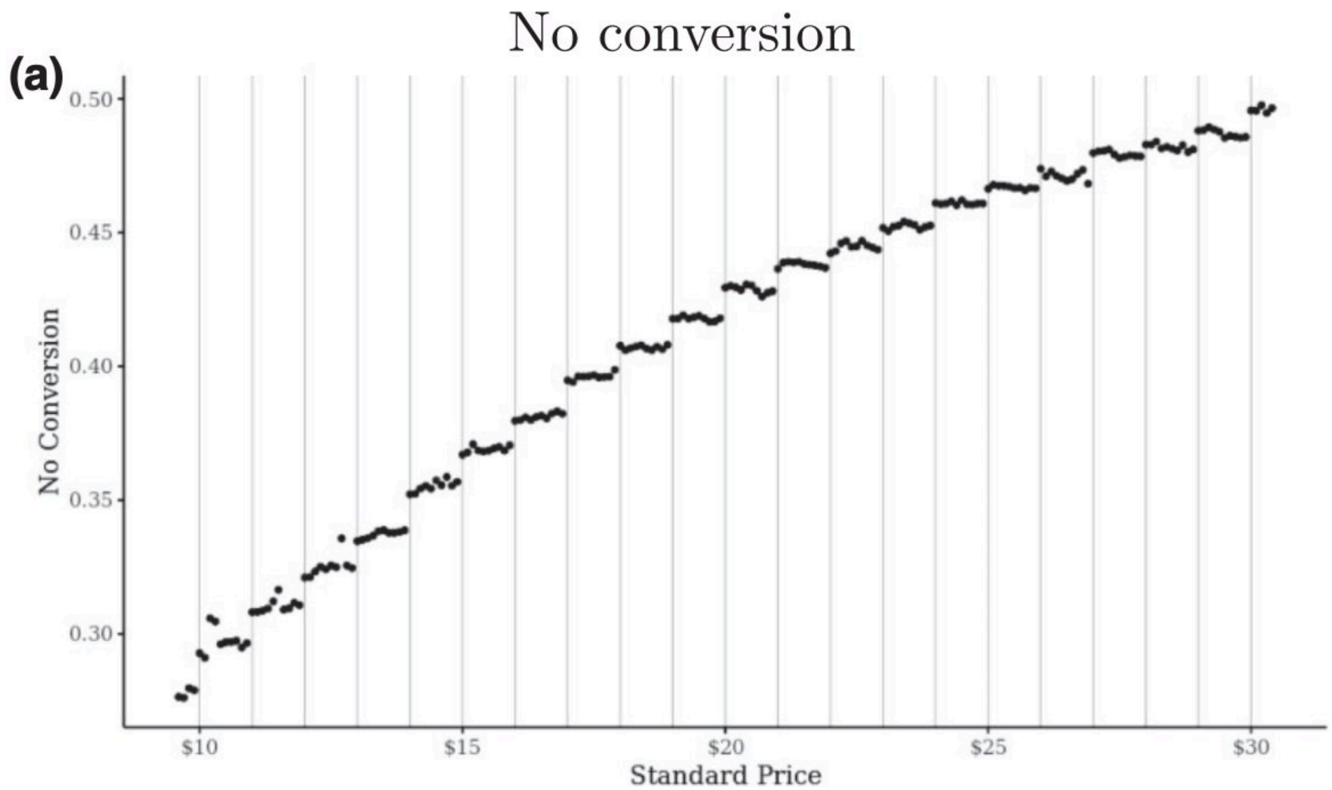


FIGURE 2
Demand curve of thirty-seven products, exhibiting drops at dollar digits

> Strulov-Shlain used ground coffee data to estimate price elasticity within each ten-cent range of prices observed in the category, showing demand discontinuities at dollar boundaries. But maybe the shoppers just hadn't had their coffee yet?

Strulov-Shlain (2023)

Left-digit bias: Lyft rides



Lyft ran a huge pricing experiment with 21+ million riders. Offer acceptances jumped discontinuously at dollar thresholds. (But why did they draw their demand curve upside down?!)

List et al. (2023)

Human factors: Price salience

- Show the price early, late or never?
  - → Drinks in a loud nightclub
  - → USPS “Forever Stamps”
  - → Price advertising, coupons
- Price salience emphasizes Savings or Exclusivity

> “If you have to ask, you can’t afford it.” When should prices be salient (upfront, hard-to-miss, repeated) vs. hidden (revealed late, easy to miss, costly to obtain)?

Human factors: Anchoring

- “You are lying on the beach on a hot lazy afternoon. For about an hour now, you have been thinking about an ice-cold bottle of your favorite beer. One of your friends gets up to make a phone call and offers to get you your favorite beer from a *small run-down grocery store* on the way back. Your friend says that the beer might be expensive and asks the maximum price that you are willing to pay. If the price is higher, your friend won’t buy the beer. What is your maximum price?
- … *fancy resort hotel* …

- Your turn:
  - → You’d like a fancy boba. The tea shop on campus sells them for $11. The location on Balboa is selling them for $3 today. Do you drive to Balboa?
  - → You’d like a Yeti cooler. Target on campus is selling them for $118. The location on Balboa is selling them for $110 today. Do you drive to Balboa?

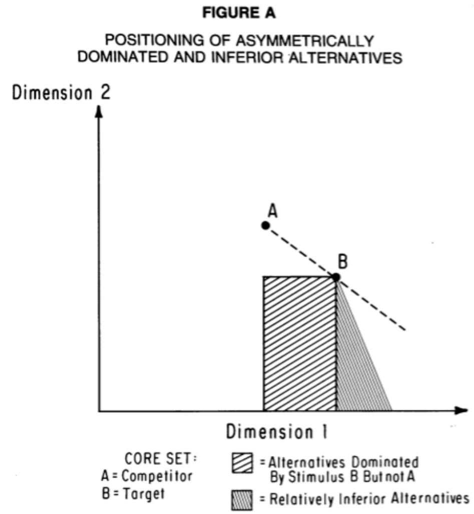
> Where do people get their reference prices? And when does bargain hunting make us happy?

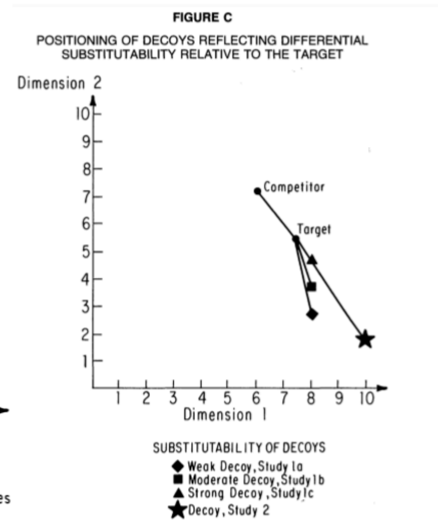
Classic study by Thaler (1985) in Marketing Science

Decoy Effects

- Brand A: Rated 50/100, priced at 1.80
- Brand B: Rated 70/100, priced at 2.60
- 33% chose A

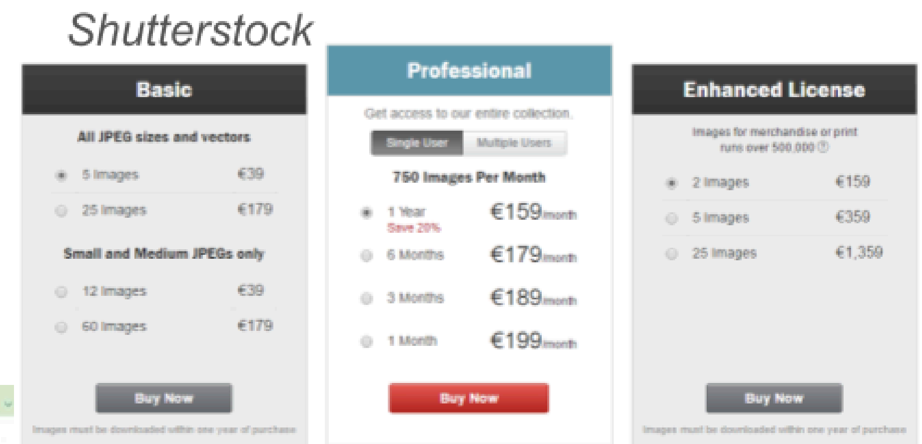
- Brand A: Rated 40/100, priced at 1.60
- Brand B: Rated 50/100, priced at 1.80
- Brand C: Rated 70/100, priced at 2.60
- 47% chose B (why?)

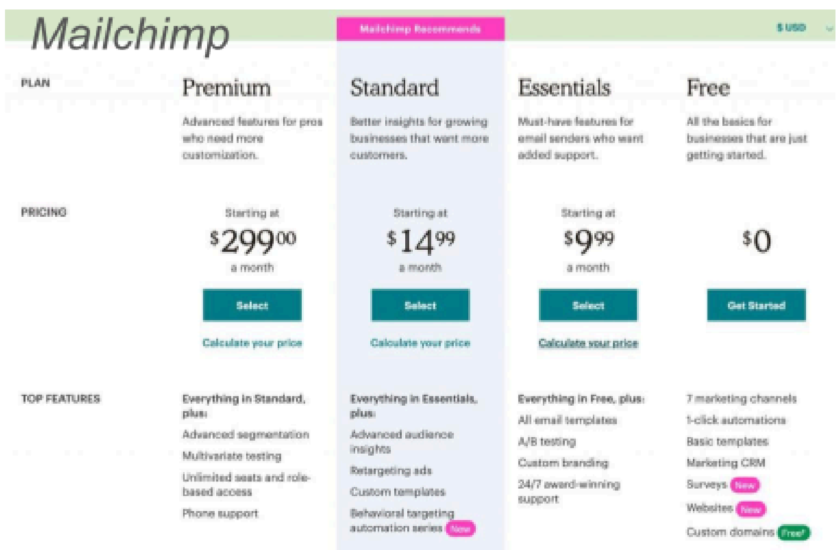




Proportionality: New items take share from existing items in proportion to their original shares (MNL: IIA property)
Substitutability: New item takes share disproportionately from more similar items (Heterogeneous Logit)
Attraction: New item may increase the relative desirability of similar items, especially within contextual evaluations (decoy)
All three effects may operate simultaneously to explain customer purchase data

Huber & Puto (1983)





Decoy pricing is common in some industries, especially Software-as-a-Service (SaaS). Sometimes price menus become cluttered

When do Attraction/Decoy Effects Obtain?

- “Ninety-one attempts to produce an attraction effect produced only 11 reliable effects.” –Yang and Lynn (2014)
- “…we are aware of only five studies that report an attraction effect using choice stimuli that are not highly stylized…we could not replicate the results of any of these studies…” Authors conducted 27 more studies in which attributes could be sensorially experienced, finding 0 decoy/attraction effects. – Frederick et al. (2014)
- When a product menu includes dominated options, consumers mistrust the choice architect and reduce purchase incidence – Bogard et al. (2024)
- Large-scale field evidence: Wu & Cosguner (2020) found significant dominant/dominated effects in calibrated model-based simulations of online diamond sales; Devine et al. (2025) found a robust but small effect of 1% in retail wine sales; Rafai et al. (2021) found no evidence when experimentally adding dominated options to online flight search results

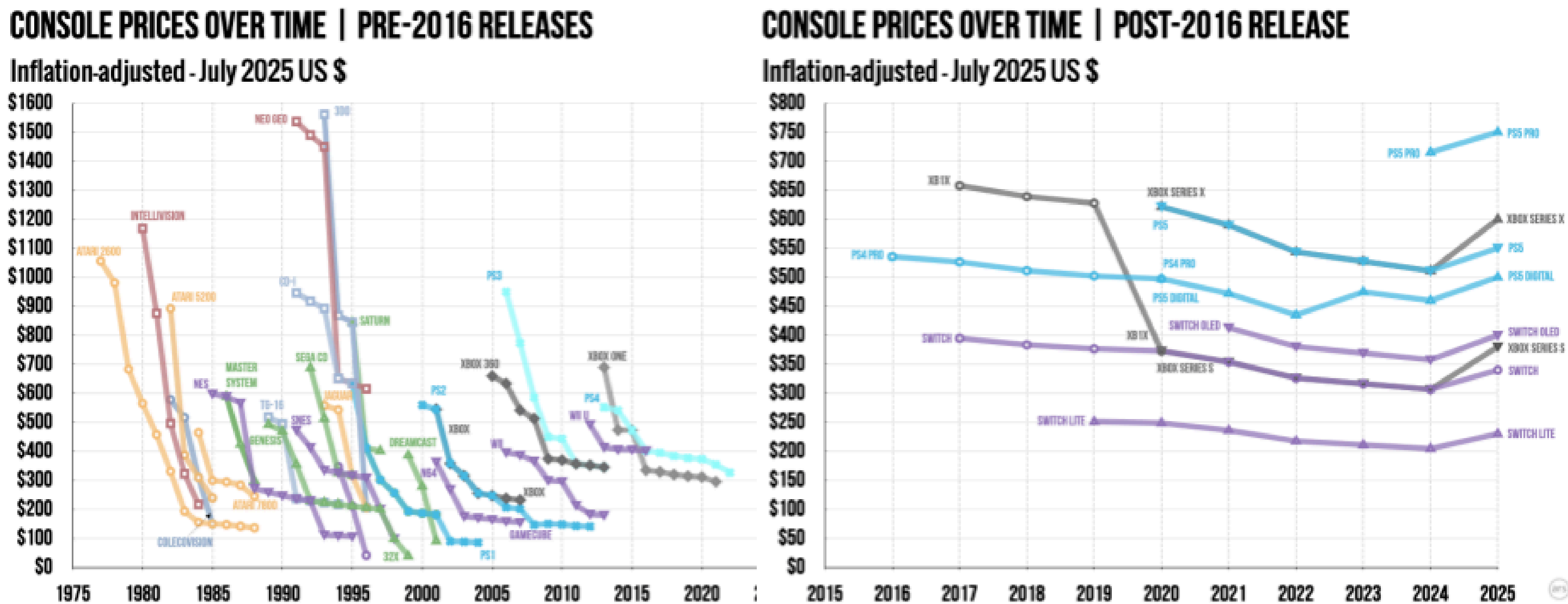
> Decoy pricing can work, but it has been overextended beyond its original boundaries of clear dominant/dominated relationships. Be careful implementing decoy pricing outside of vertical product differentiation, and never ever implement without careful testing

Yang and Lynn (2014) | Frederick et al. (2014) | Bogard et al. (2024) | Wu & Cosguner (2020) | Devine et al. (2025) | Rafai et al. (2021)

Economic factors: Price discrimination

- Amazon v. B&N
- Purchase time: Airline, Cruise tickets
- Needs: e.g. Business vs. Home segments
- Skimming by delivery time: Movie release windows
- Geography: Typically accounts for 20% of variation in online prices
- Quantity: Cups of coffee, Paper towels
- New/loyal customer, merit (veterans, seniors), ability to pay/sliding scale, value provided, cost of supply

> Why set only one price when you can set many prices instead? Always frame differences as discounts: a “student discount” sounds much nicer than the equivalent “non-student premium.” Framing is more acceptable when it advantages a vulnerable segment, without explicitly excluding another segment



> Game consoles used to use price skimming to price discriminate between gamers, but that appears to have stopped. Why?

Ars Technica

Economic factors: Beware a price war!

- If you explicitly mention a competitor's price
  - → You make Customer aware of Competitor
  - → Competitor may notice: You invite them to match or retaliate
- Better to price-compare vs. unnamed/generic competitor
- Who wins a price war?
  - → Only one winner: Customer
  - → All firms suffer, some die
  - → Most likely to survive: Seller with lowest cost structure
  - → Smart firms avoid price wars & keep costs secret

> Why does cost secrecy usually help to avoid price wars?

Price Elasticity of Demand

- $elas. = \frac{d(lnq)}{d(lnp)} = \frac{p}{q}\frac{dq}{dp} \leq 0$; Typically becomes more negative as price rises
- Elasticity is “scale-free” : % change response to % change, because a change in ln(X) is approximately equal to the % change in X
- For $-1 < elas. < 0$, demand is inelastic; for $elas. < -1$, demand is elastic
- Elasticity calculation depends on method
  - → We can approximate $\frac{dq}{dp}$ with two points and no demand model
  - → If we specify demand model, we can calculate $\frac{dq}{dp}$ based on the model
  - → Elasticity calculations will be sensitive to interval widths and model assumptions

> Revenue is maximized at $elas. = -1$. For $elas. < -1$, you can increase revenue by reducing price. For $elas. > -1$, you can increase revenue by increasing price. Profits are maximized at $MR = MC$

Price Elasticity of Demand

- $elas. = \frac{d(\ln q)}{d(\ln p)} = \frac{p}{q}\frac{dq}{dp}$
- A special class of demand functions have constant elasticity
  - → $q = e^{\alpha} p^{\beta}$ for $\alpha > 0$ & $\beta < -1$, then $elast. = \beta$
  - → Implies $\ln q = \alpha + \beta \ln p$, called "log-log"
  - → Need exogenous price variation to estimate a causal effect β; otherwise, β should be interpreted as a correlation
- C.E. imposes a restrictive shape on demand & enables easy price optimization, given marginal cost data, but if C.E. shape is incorrect, C.E. approach will lead to suboptimal pricing

> C.E. demand is popular because it is easy. However, C.E. assumes that customers with different wtp all have the same price sensitivity. When might that make sense, or not make sense?

Price optimization: CED vs. het. MNL

Constant Elasticity of Demand

- Assume the convenient function:
  → $q(p) = e^{\alpha} p^{\beta}$
- This implies:
  → $\ln(q) = \alpha + \beta \ln(p)$
- Slope coefficient is elasticity:
  → $\frac{d \ln(q)}{d \ln(p)} = \frac{p}{q} \frac{dq}{dp} = \beta$
- You can show that maximizing $\pi = (p - c)q(p)$ yields optimal price p^*:
  → $p^* = \frac{c}{1+\frac{1}{\beta}}$, requires $\beta < -1$ (elastic demand)

Het. MNL Demand Model

- Estimated sales from market-share predictions at each candidate price:
  → $q(p) = M\hat{s}(p)$
- Profit at each candidate price:
  → $\pi(p) = q(p)(p - c)$
- Try many candidate prices, pick the one that maximizes profit:
  → $p^* = \arg\max_{\{p\}} \pi(p)$

RHS is called a "grid search." It's less mathematically elegant but more robust, and generally works better. You can use a grid search to evaluate whether demand elasticity is constant (how?); but you cannot use CE Demand to predict the results of a grid search

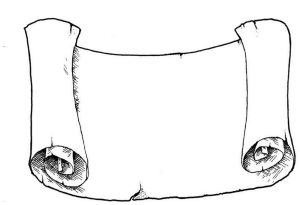
Best-response pricing curve

- If a rival raises its price, some of its customers will choose to buy from us. If it cuts its price, some of our customers will choose to buy from the rival. Either change shifts our demand curve.
- When our demand curve shifts, our optimal profit-maximizing price shifts in the same direction. If demand shifts up, we should increase price; if demand shifts down, we should decrease price.
- The best-response pricing curve graphs that logic, based on our profit-maximization calculations. It shows how our optimal price p* changes as a function of our competitor's price.

Why does the best-response pricing curve slope up?

Class script

- Use demand model to trace out a demand curve
- Compare different arc elasticity results
- Conduct a grid search to find the profit-maximizing price, all else constant
- Compare the grid search result to the CE-demand price
- Consider multi-product price optimization



Wrapping up

Competition

- Calculate Apple's best-response A1 price as a function of Samsung's S1 price
  - → Use the class dataset with `mdat1` and the week 6 model `out10`; assume Apple's A1 marginal cost is $450
  - → For each S1 price in {$599, $649, $699, $749, $799, $849, $899, $949, $999}, grid-search A1 prices and find the A1 price that maximizes Apple's A1 profit, holding all other prices fixed
- Plot A1's profit-maximizing price (y-axis) against S1's price (x-axis, using the 9 S1 price points above) — this is Apple's best-response pricing curve

Recap

- The most common and easiest price setting methods are competitor price matching and cost-based pricing. Both are incomplete
- Consumers usually expect product prices to reflect quality positions in the marketplace
- Optimal pricing requires attention to both economic factors and human factors

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Going further

- Willingness to Pay Measurement Approaches
- Science of price experimentation at Amazon
- Behavioral Pricing
- More than a Penny's Worth: Left-Digit Bias and Firm Pricing
- Dynamic Online Pricing with Incomplete Information Using Multiarmed Bandit Experiments
- Universal Paperclips : Fun price setting game